

Article

Development of a Rehabilitation Chair Design Based on a Functional Technology Matrix and Multilevel Evaluation Methods

Xinyan Yang¹ and Zhongfeng Zhang^{1,2,*}¹ College of Furniture and Art Design, Central South University of Forestry Technology, Changsha 410004, China; 18919104575@163.com² Green Furniture Engineering Technology Research Center, State Forestry and Grassland Administration, Changsha 410004, China

* Correspondence: t19990735@csuft.edu.cn; Tel.: +86-158-2280-3305

Featured Application: Using Structural Design to Reduce Product Development Costs and Increase Patient Active Rehabilitation Motivation.

Abstract: In order to meet the diversified rehabilitation needs of people, it is necessary to design a product for the active rehabilitation of patients. Existing rehabilitation chairs use intelligent massage, which can cause problems such as large massage areas, inability to massage locally, large chair size, and inability to meet the continuous use of the damaged parts. In this paper, the modular design method and multi-layer evaluation method are used to solve the problems related to rehabilitation chairs. The authors use the questionnaire survey method and the functional technology matrix method to determine the functional requirements of the rehabilitation chair, and then use the multilevel evaluation methods, including the AHP method, entropy weight method, and grey correlation analysis, to optimize the functional solutions of the rehabilitation chair, and finally obtain a chair for the rehabilitation of patients with upper and lower limb disorders. Problems such as the generalization of rehabilitation scope and non-durable use of components were solved, and the purpose of active exercise was achieved. This study verifies that the use of the multilevel decision evaluation method can effectively improve the efficiency of program decision-making and provides a theoretical and practical basis for the design of similar products.

Keywords: rehabilitation chair design; functional technology matrix; multilevel evaluation methodology

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1. Introduction

Limb movement dysfunction is one of the most common rehabilitation problems in patients with stroke, hemiplegia, ageing, and chronic diseases [1]. It is estimated that by 2047, older people will outnumber children globally [2]. Research and improvement of rehabilitation products can improve the quality of rehabilitation for people with physical disabilities. Currently, most of the product designs related to the rehabilitation of physical disabilities are exoskeleton devices or positioning and massage rehabilitation robots [3,4]. The design of rehabilitation robots began in the 1980s, and MIT-Manus was one of the first organizations to develop rehabilitation robots, which are important for achieving functional compensation and reconstruction for patients with disabilities [5–9]. The process of limb rehabilitation is a long and slow process, and existing rehabilitation products mainly focus on localized limb rehabilitation and cannot provide full-cycle and multi-stage rehabilitation use. Due to the complexity of the upper and lower limb interaction structure, products that realize the simultaneous rehabilitation of the upper and lower limbs will inevitably increase the cost, which necessitates the use of the concepts of sustainable development and circular economy to manufacture rehabilitation products [10]. Dividing a complex product system

into different types of components and then considering their functions, manufacturing, and interworking processes has the advantage of reducing the complexity of product development and improving resource utilization. In conclusion, this study presents a rehabilitation chair design that aims to promote products for the active rehabilitation of patients with upper and lower limb dysfunction by promoting self-rehabilitation in physically disabled people.

The remaining sections of this paper are structured as follows. In Section 2, the authors briefly review the literature on the methodology of the study, the shortcomings of previous studies, and areas for improvement and innovation in this research area. In Section 3, the authors introduce the research framework for the design of the rehabilitation chair and systematically discuss the design process of each functional structure of the rehabilitation chair. Section 4 demonstrates the specific structure and use of the final preferred solution for the rehabilitation chair and discusses the methodology and experience used for the modular rehabilitation chair. Section 5 is the concluding section which summarises the innovations in the design of the modular rehabilitation chair and provides some ideas for future research directions to extend the scope of this study.

2. Materials and Methods

2.1. Design Methodology

The Analytic Hierarchical Process (AHP) is a system analysis method proposed by American operations researcher Satty in the early 1970s [11]. It decomposes complex problems into constituent units and groups these elements into orderly progressive hierarchical structures according to the dominant relationship, i.e., it establishes a hierarchical analysis model and then determines the relative importance of each element by comparing two by two, so as to complete the quantitative assessment of qualitative indicators. The earliest entropy theory was proposed by the German physicist Clausius in 1856. “Entropy” is a measure of the degree of disorder in the system. If the information entropy of the indicator is smaller, the more information the indicator provides, the greater the role in the comprehensive evaluation and the higher the weight. Combining entropy weight theory with the hierarchical analysis method can make up for the subjectivity in comprehensive evaluation [12].

Grey Relational Analysis (GRA) can eliminate the weaknesses of fuzzy or stochastic methods. The strength of GRA is that it can deal with concepts with fuzzy internal information and clear external information, while the strength of fuzzy and stochastic methods is that they can deal with concepts with precise internal information and fuzzy external information [13].

The functional technology matrix method achieves concretization of the design object step by step through the system synthesis of the product, functional analysis, and analysis of technical elements. First of all, screen the function, disassemble the total function into sub-functions, and then use the function matrix, i.e., a qualitative or quantitative method, to analyze the different combination forms of realizing the technical way of sub-functions, and finally seek a feasible design structure or scheme.

This study uses the questionnaire survey method to obtain the functions of the rehabilitation chair and establish a functional technology matrix after screening. By analyzing the different combination forms of functional technology paths, feasible functional solutions are determined. Finally, multi-level fuzzy comprehensive evaluation methods such as the AHP method, entropy weight method, and grey correlation analysis method are combined to determine the final scheme of the rehabilitation chair [14–18]. The specific steps are shown in Figure 1.

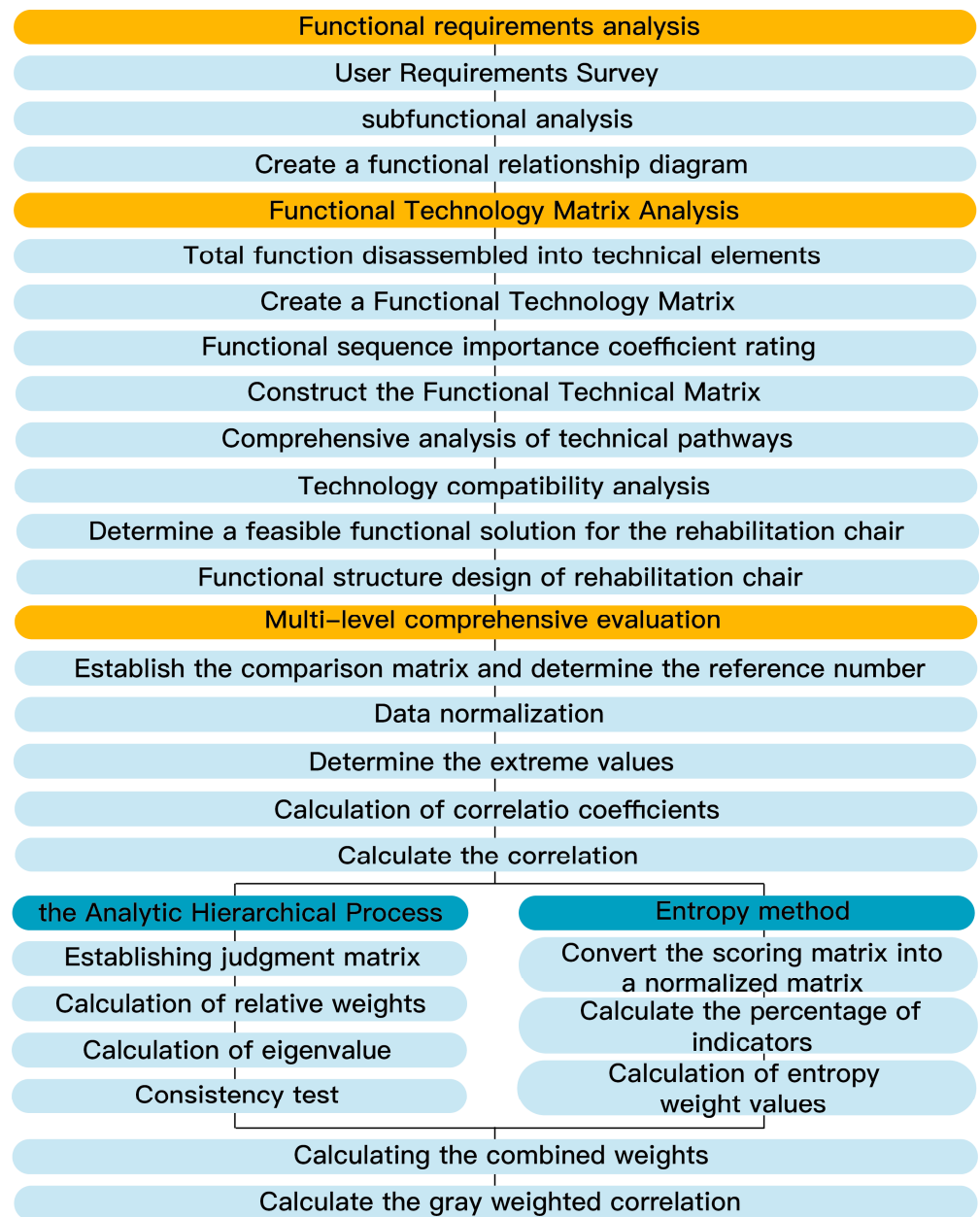


Figure 1. Rehabilitation Chair Design Process.

2.2. Design Process

2.2.1. Functional Requirement Analysis of Rehabilitation Chairs

The authors conducted observation and questionnaire research on physically disabled groups in three nursing homes in Changsha, Hunan Province, namely Taikang Home Xiangyuan, Poly Changsha Tianxin Yuehui, and Changsha Puzhen Nursing Home for the Elderly, to collect the current demand for rehabilitation chairs for people with physical disabilities. The questionnaire was designed to include four aspects: 1. Whether rehabilitation chairs are needed for daily rehabilitation training; 2. How long do the people with physical disabilities use the chairs every day; 3. What rehabilitation functions are needed to meet the stage of rehabilitation needs; 4. How the functional modules can be laid out to effectively reduce costs. This research issued 300 questionnaires. A total of 300 were recovered, including 298 valid questionnaires, displaying an effective rate of 99.3%. After collating the results of the questionnaires, as shown in Figure 2, the main functions of upper limb rehabilitation are back massage, stretching, chest expansion, arm

raising, and grasping; the main functions of lower limb rehabilitation are flexion, stretching, leg lifting, foot sports, and leg tapping.

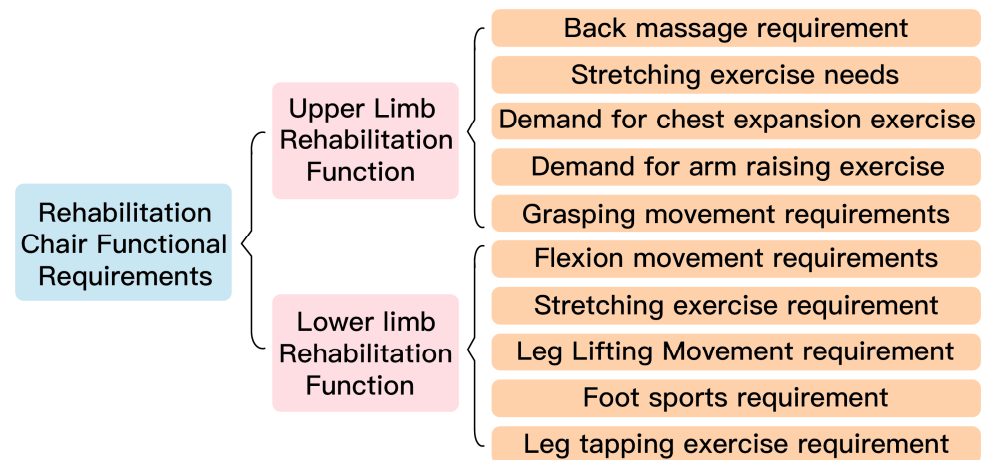


Figure 2. Rehabilitation Chair Functional Requirements.

2.2.2. Establishing the Functional Technology Matrix of the Rehabilitation Chair

Decompose the overall function of the rehabilitation chair until it is decomposed into functional elements composed of structures or components, establish the functional technology matrix diagram with the decomposed technical elements, comprehensively analyze the technical pathways of each function, and screen the functional solutions with high practical value, sustainable use, superior technology and economy, and aesthetics. The specific steps are as follows.

Step 1: Total function disassembled into technical elements.

Organize the functional elements according to the realization relationship between the functions. Distinguish between basic and non-basic functions according to the importance of the functions and decompose the functions into technical elements according to the purpose of rehabilitation and means of realization, as shown in Figure 3.

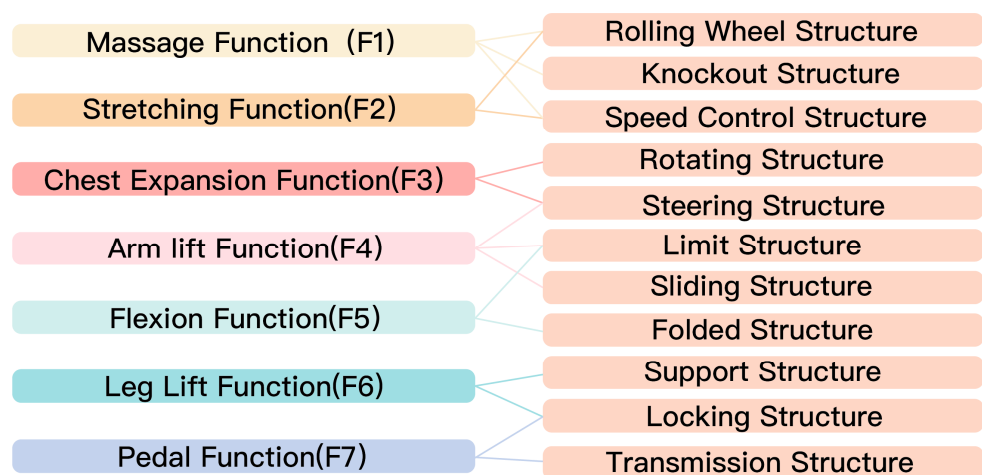


Figure 3. Technology dismantling in functionality.

Step 2: Create a Functional Technology Matrix

As shown in Table 1, a functional matrix is a combination of different forms of assistive technology around a product's functions, qualitatively or quantitatively analyzing the different forms of achieving each function to find a reliable solution.

Table 1. Functional technology matrix (example).

Functions		Technology Pathways			
F1 ¹	T1(1)	T2(2)	T3(3)	...	T1(m)
F2	T2(1)	T2(2)	T2(3)	...	T2(m)
F3	T3(1)	T3(2)	T3(3)	...	T3(m) ²
...	...	...	...	...	...
F _n 1	T _n (1)	T _n (2)	T _n (3)	...	T _n (m) ³

¹ F for Function; F_n stands for the nth function; ² T for technology; m for m technologies in each function; ³ T_n(m) represents the mth technology in the nth function.

Step 3: Functional sequence importance coefficient rating

The functional importance coefficient is also known as the functional coefficient or functional index. The rehabilitation chair functional importance coefficient rating is based on the different roles of each functional element in the whole system, through the mandatory scoring method (0–1 or 0–4 scoring method), multiple proportional scoring method, logical scoring method, ring scoring method, and so on. In this paper, the 0–4 scoring method is adopted, that is, 10 professionals are asked to compare the two functional elements, and the specific scoring is shown in Table 2, where the very important functional element scores 4 points, and the other very important functional element scores 1 point; the more important functional element scores 3 points, and the other not-so-important functional element scores 1 point; when they are of equal importance or basically of equal importance, each of the two functional elements scores 2 points, the very unimportant scores 1 point, and no points are scored for their own comparisons.

Table 2. Functional Sequence Importance Coefficient Rating Scale.

Evaluation Mark	Description of the Meaning of the Score
4	Extremely important
3	Comparatively important
2	Equally important
1	Not too important or compared to itself
0	Unimportant

According to the example Table 1 to establish the functional technology matrix, we get the functional results of the rehabilitation instrument as shown in Table 3, specifically: the massage function is 0.174, the stretching function is 0.130, the chest expansion function is 0.081, the arm lifting function is 0.174, the flexion and extension function is 0.093, the leg patting function is 0.093, and the foot stirrup function is 0.174.

Table 3. Rehabilitation Chair Functional Elements 0–4 Rating Scale.

Functions	Function Comparison Score							Function Importance Score	Functional Importance Factor
	F1	F2	F3	F4	F5	F6	F7		
F1	0	3	2	3	2	3	2	15	0.174
F2	1	0	1	3	1	2	3	11	0.130
F3	2	3	0	3	2	3	2	15	0.174
F4	1	1	1	0	1	2	1	7	0.081
F5	2	3	2	3	0	3	2	15	0.174
F6	1	2	1	2	1	0	1	8	0.093
F7	2	1	2	3	2	3	0	15	0.174
Total								86	1

Step 4: Construct the Functional Technical Matrix

Based on the results of the rating of functional importance factors, the functional technology matrix was constructed. In order to expand the scope of the technology path-

way and improve the creativity and effectiveness of the final solution, the latest scientific research results and manufacturing technologies were adopted as the technology pathway for the rehabilitation chair. Table 4 shows the functional technology path of the rehabilitation chair, in which the massage function mainly utilizes structures such as rollers, worm gear structures, and struts, the extension function mainly utilizes structures such as rollers, socketed telescopic sliders, and telescopic sliders, the chest expansion function mainly utilizes structures such as screws, chucks, mandrels, and telescopic rods, the arm lift function mainly utilizes structures such as supports, rotating sections, and rotating telescopic sections, and the flexion function mainly utilizes folding rod ends, locking rings, and fixing pins, the leg knocking function mainly utilizes wooden knockers, support wheels, and hinged rods, and the pedal function mainly utilizes turntables, connecting plates, and foot pedals.

Table 4. Rehabilitation chair functional technology matrix.

Functions	Technology Pathways		
F1 ¹	T1(1) ²	T1(2)	T1(3)
	Roller	Worm gear	Struts
F2	T2(1)	T2(2)	T2(3)
	Roller	Sleeve type expansion slider	Telescopic slider
F3	T3(1)	T3(2)	T3(3)
	Screw	Mandrel	chuck
F4	T4(1)	T4(2)	T4(3)
	Support piece	Rotating section	rotating telescopic part
F5	T5(1)	T5(2)	T5(3)
	Folded movable rod end	Locking ring	Retaining pin
F6	T6(1)	T6(2)	T6(3)
	Wooden knockout head	Support wheel	Hinge rod
F7	T7(1)	T7(2)	T7(3)
	Turntable	Connection plate	Foot pedal

¹ F1 represents the first function. ² T1(1) represents the first technology in function 1.

Step 5: Comprehensive analysis of technical pathways

Comprehensive analysis of the technical pathways refers to the comprehensive consideration of the indicators of the evaluation of the rehabilitation chair program, such as technical, practical, aesthetic, and other indicators. The scoring items are developed according to the requirements, and the scoring method refers to the 0–4 scoring method. For example, when two technical elements, T1 and T2, are compared, if T1 is more important, then T1 receives a score of 3 and T2 receives a score of 1; if T1 is very important, then T1 receives a score of 4 and T2 receives a score of 0; and if the two elements are equally important, then both elements receive a score of 2; refer to the scoring method in Table 2. Finally, the scores obtained for each technical element are added together and then divided by the total score of technical elements, see formula (1) to obtain the Technical Index Factors.

The evaluation indexes of the functional elements of the rehabilitation chairs in this study include technology, environmental friendliness, and applicability, and the technical elements are scored with each index as an evaluation dimension, where Tables 5–7 are obtained.

$$Z = \frac{1}{N} \sum_{n=1} T_n \quad (1)$$

Z is the technical indicator coefficient. n is the number of technical elements in the function.

Table 5. Evaluation of the technicality index coefficients of the functional elements of the rehabilitation chair.

Functions		Technology Element Score			Technical Index Scores	Technical Index Factors
F1	T1(1)	T1(1)	T1(2)	T1(3)		
		0	2	3	5	0.42
	T1(2)	2	0	3	5	0.42
F2	T1(3)	1	1	0	2	0.17
	T2(1)	T2(1)	T2(2)	T2(3)		
		0	1	1	2	0.17
F3	T2(2)	3	0	2	5	0.17
	T2(3)	3	2	0	5	0.42
	T3(1)	T3(1)	T3(2)	T3(3)		
F4	T3(1)	0	2	3	5	0.42
	T3(2)	2	0	3	5	0.42
	T3(3)	1	1	0	2	0.17
F5	T4(1)	T4(1)	T4(2)	T4(3)		
		0	3	3	6	0.46
	T4(2)	2	0	3	5	0.38
F6	T4(3)	1	1	0	2	0.15
	T5(1)	T5(1)	T5(2)	T5(3)		
		0	2	2	4	0.33
F7	T5(2)	2	0	3	5	0.42
	T5(3)	2	1	0	3	0.25
	T6(1)	T6(1)	T6(2)	T6(3)		
F8	T6(1)	0	2	1	3	0.25
	T6(2)	2	0	2	4	0.33
	T6(3)	1	2	0	5	0.42
F9	T7(1)	T7(1)	T7(2)	T7(3)		
		0	2	2	4	0.25
	T7(2)	2	0	2	4	0.25
F10	T7(3)	2	2	0	4	0.25

Table 6. Evaluation of environmental protection index coefficients of the functional elements of the rehabilitation chair.

Functions		Technology Element Score			Technical Index Scores	Technical Index Factors
F1	T1(1)	T1(1)	T1(2)	T1(3)		
		0	2	1	3	0.25
	T1(2)	2	0	1	3	0.25
F2	T1(3)	3	3	0	6	0.50
	T2(1)	T2(1)	T2(2)	T2(3)		
		0	3	1	4	0.33
F3	T2(2)	1	0	1	2	0.17
	T2(3)	3	3	0	6	0.50
	T3(1)	T3(1)	T3(2)	T3(3)		
F4	T3(1)	0	2	3	5	0.42
	T3(2)	2	0	3	5	0.42
	T3(3)	1	1	0	2	0.16
F5	T4(1)	T4(1)	T4(2)	T4(3)		
		0	2	3	5	0.42
	T4(2)	2	0	3	5	0.42
F6	T4(3)	1	1	0	2	0.16
	T5(1)	T5(1)	T5(2)	T5(3)		
		0	3	1	4	0.33
F7	T5(2)	1	0	1	2	0.17
	T5(3)	3	3	0	6	0.50
	T6(1)	T6(1)	T6(2)	T6(3)		
F8	T6(1)	0	2	3	5	0.42
	T6(2)	2	0	3	5	0.42
	T6(3)	1	1	0	2	0.16
F9	T7(1)	T7(1)	T7(2)	T7(3)		
		0	3	2	5	0.42
	T7(2)	1	0	1	2	0.16
F10	T7(3)	2	3	0	5	0.42

Table 7. Evaluation of applicability index coefficients of the functional elements of the modular rehabilitation chair.

Functions		Technology Element Score			Technical Index Scores	Technical Index Factors
F1	T1(1)	T1(1)	T1(2)	T1(3)		
	T1(1)	0	2	3	5	0.42
	T1(2)	2	0	3	5	0.42
F2	T1(3)	1	1	0	2	0.16
	T2(1)	T2(1)	T2(2)	T2(3)		
	T2(1)	0	1	1	2	0.17
F3	T2(2)	3	0	3	6	0.50
	T2(3)	3	1	0	4	0.33
	T3(1)	T3(1)	T3(2)	T3(3)		
F4	T3(1)	0	1	3	4	0.33
	T3(2)	3	0	3	6	0.50
	T3(3)	1	1	0	2	0.17
F5	T4(1)	T4(1)	T4(2)	T4(3)		
	T4(1)	0	3	3	6	0.50
	T4(2)	1	0	3	4	0.33
F6	T4(3)	1	1	0	2	0.17
	T5(1)	T5(1)	T5(2)	T5(3)		
	T5(1)	0	3	2	5	0.45
F7	T5(2)	1	0	1	2	0.18
	T5(3)	2	3	0	4	0.36
	T6(1)	T6(1)	T6(2)	T6(3)		
F8	T6(1)	0	1	2	3	0.23
	T6(2)	3	0	2	5	0.38
	T6(3)	3	2	0	5	0.38
F9	T7(1)	T7(1)	T7(2)	T7(3)		
	T7(1)	0	3	2	5	0.42
	T7(2)	1	0	1	2	0.16
F10	T7(3)	2	3	0	5	0.42

Step 6: Technology compatibility analysis

Technology compatibility analysis is an important method to determine whether the combination of technology paths of each functional element is compatible with each other and the feasibility of the combination scheme. The $N \times N$ order matrix is constructed using the adjacency matrix method to perform the technical pathway compatibility analysis, as shown in Table 8. If the combination condition is feasible, it is counted as 1, and if the combination condition is not feasible, it is counted as 0. The technical pathway ratings of T1(1), T1(2), T1(3) ..., etc., are performed sequentially, and finally a number of combination schemes Q are obtained.

Table 8. Technical compatibility scoring matrix.

	Compatibility			
	T1(1)	T1(2)	...	Tn(m)
T1(1)	0/1 ²	0/1	...	0/1
T1(2)	0/1	0/1	...	0/1
...	...	...	...	...
Tn(m) ¹	0/1	0/1	...	0/1

¹ Tn(m) represents the mth technology in the nth function. ² 1 for compatibility, 0 for non-compatibility.

$$Q = \{q_1, q_2, q_n, \dots, q_m\} \quad (2)$$

Here, q_n is the n th feasible solution and m is the number of feasible solutions.

Step 7: Determine a feasible functional solution for the rehabilitation chair.

As shown in Table 8, the pathways with higher technical excellence index coefficient scores in Tables 5–7 were subjected to technical compatibility matrix analysis, with compatibility as 1 and incompatibility as 0. The feasible options for the rehabilitation chair function were obtained as follows:

- (1) Massage Function: roller, worm gear worm gear, strut.
- (2) Stretching Function: sleeve type expansion slider, telescopic slider.
- (3) Chest Expansion Function: screw, mandrel.
- (4) Arm Lift Function: support piece, rotating section.
- (5) Flexion Function: folded movable rod end, retaining pin.
- (6) Leg Lift Function: wooden knockout head, hinge rod.
- (7) Pedal Function: turntable, foot pedal.

Step 8: Functional structure design of the rehabilitation chair

According to the above technical elements, combined with the functions obtained from user research, including back massage, leg flexion and extension, chest expansion movement, arm lifting, leg bending, leg lifting, and footrest movement, the rehabilitation chair functional program is designed. This is shown in Figures 4–6.

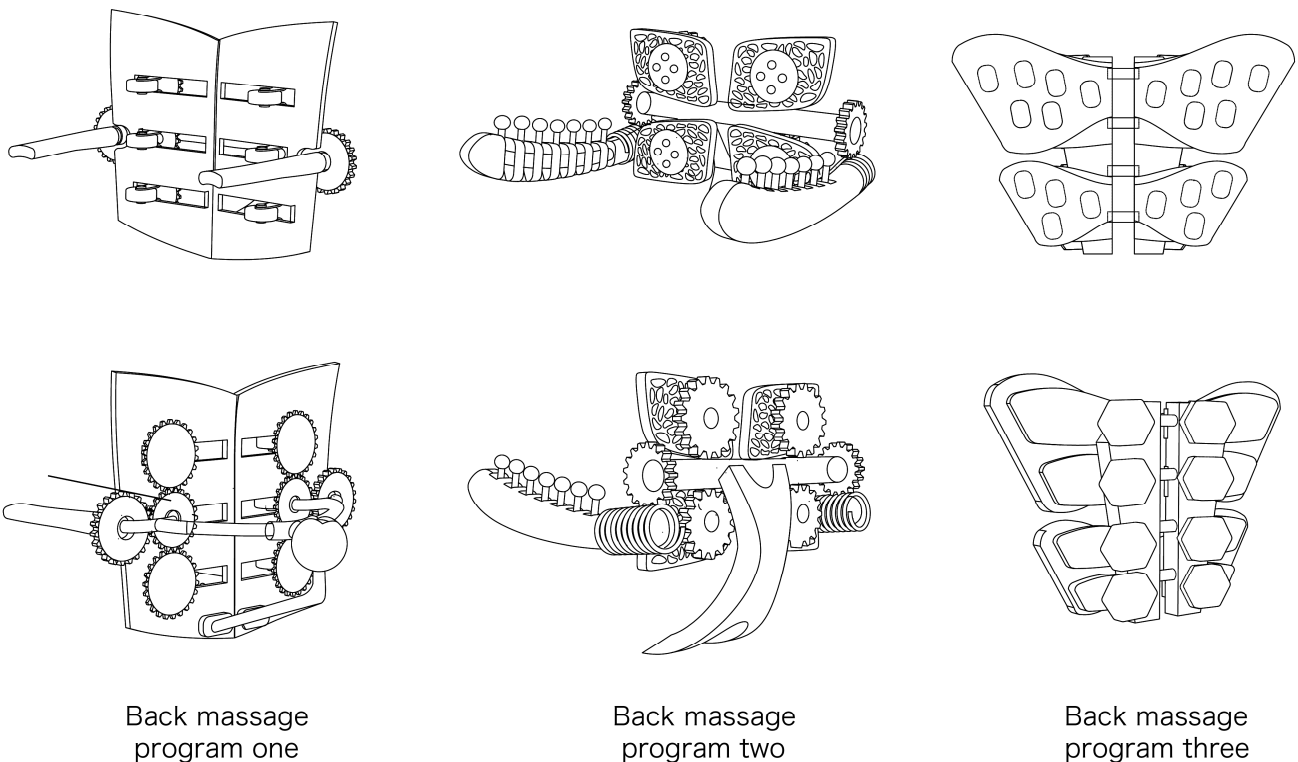


Figure 4. Backrest massage program.

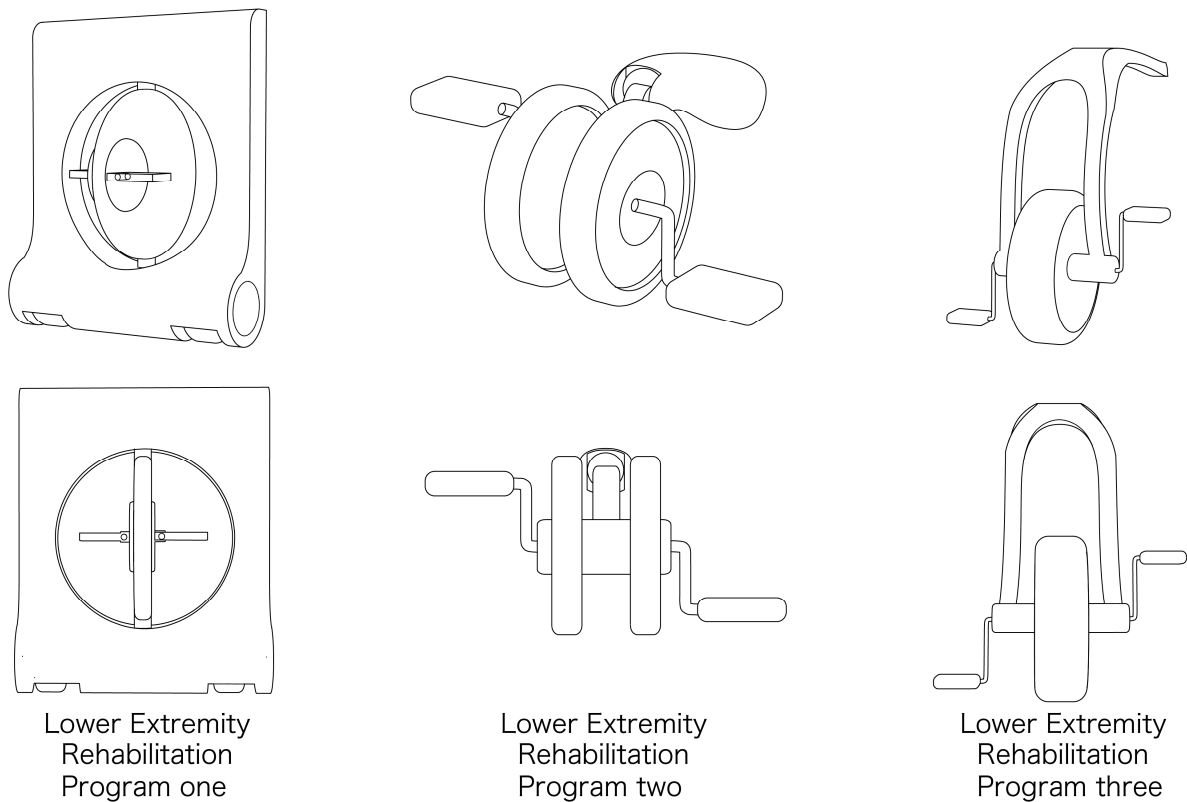


Figure 5. Lower extremity rehabilitation program.

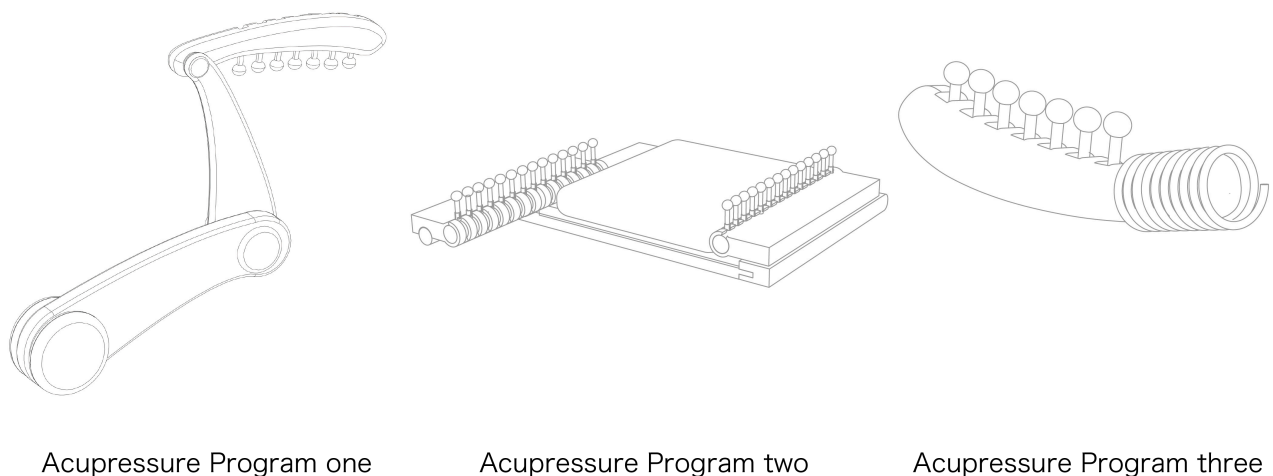


Figure 6. Acupressure program.

2.2.3. Multi-Level Comprehensive Evaluation

Using the gray correlation method to get the correlation degree of each functional scheme and then using the AHP-Entropy weight method to calculate the comprehensive weight, it is possible to finally calculate the gray weighting weight of each scheme to obtain the optimal scheme. This process is shown in Figure 1.

Step 1: Establish the comparison matrix and determine the reference number.

Establish the decision-making evaluation system of the rehabilitation chair program, as shown in Figure 7, the evaluated program is C ($C = 1, 2, 3, n$), the set of evaluation level 1 indicators is $E' = (E'_1)$, and the set of level 2 indicators is $E' = E'_{1m}, E'_{2m}, E'_{3m}, \dots, E'_{nm}$.

Determine the optimal value of each indicator as the reference data column, and set the reference data column as $E'_0 = E'_{01}, E'_{02}, E'_{03}, \dots, E'_{0m}$.

rehabilitation function Indicator(E1)	Structural performance indicator(E2)	Safety load Indicator(E3)
Rehabilitation Modality(E11)	Sensitivity(E21)	Structural Adaptation(E31)
Rehabilitation Effect(E12)	Structural Strength(E22)	Quality and Safety(E32)
Functional Goal(E13)	Structural impact resistance(E23)	Structural Stability(E33)
Degree of Recovery(E14)	Structural comfort(E24)	Load Capacity(E34)
Green Recycling Indicator(E4)	Operational use indicator(E5)	Emotional Interaction Indicator(E6)
Structural Redesign(E41)	Speed Control Capability(E51)	Ease of Use(E61)
Recyclability(E42)	Exercise Range(E52)	Ergonomics(E62)
Structural Interchangeability(E43)	Exercise Durability(E53)	Fault Tolerance(E63)
Material Safety(E44)	Intensity Control(E54)	Easy to Learn(E64)

Figure 7. Decision-making evaluation system for the rehabilitation chair program.

Step 2: Data normalization.

To improve the validity of the data, the different indicators are normalized. The comparison matrix $E = E_{nm} = (E'_{jk})_{nm} = (j = [1, m]; k = [1, n])$ was obtained. The normalized data for each of the three types of scenarios are shown in Figure 8.

E(backrest)	4.2	4.6	4.6	3.8	4.3	3.9	3.4	3.8	4.8	4.4	4.5	3.9	4.3	3.9	4.7	4.2	4.4	3.7	4.5	4.6	4.5	4.2	4.1	4.8
	4.6	4.7	4.9	3.8	4.6	4.7	4.0	4.2	4.5	4.5	4.6	4.7	4.7	4.2	4.9	4.1	4.5	4.2	4.2	4.4	4.4	4.8	3.8	4.5
	4.0	4.5	4.5	4.5	3.8	4.6	3.9	3.8	3.7	4.4	4.3	4.3	4.5	4.2	4.6	4.2	3.6	4.5	3.9	4.0	4.0	3.9	4.5	4.6
E(Lower limb)	4.1	4.5	4.4	4.4	4.5	4.7	4.4	4.2	4.5	4.4	4.6	4.3	4.3	4.4	4.5	4.5	4.5	3.9	4.5	3.8	4.8	4.3	4.3	4.3
	4.4	4.3	4.5	4.3	4.5	4.3	4.7	4.7	4.6	4.6	4.7	4.8	4.8	4.7	4.6	4.2	4.8	4.3	4.7	4.6	4.5	4.4	4.6	4.5
	4.0	4.2	4.4	4.5	4.6	4.8	4.7	4.6	4.4	4.5	4.0	4.4	4.1	4.4	4.2	3.7	3.7	3.8	4.4	4.3	4.5	4.3	4.5	4.6
E(Armrests)	4.0	4.3	4.4	4.4	3.8	3.8	3.9	3.7	4.0	3.9	4.1	4.0	4.2	4.0	4.6	3.8	3.8	3.9	4.2	3.9	3.8	3.7	3.7	3.9
	4.0	4.3	4.5	3.7	4.0	3.9	3.8	4.1	4.0	4.0	4.2	4.0	4.4	4.0	4.5	3.8	4.2	3.8	3.9	4.0	3.9	4.1	4.0	4.0
	3.5	3.4	3.9	3.2	3.9	3.8	3.1	3.2	4.1	4.0	4.6	4.0	4.3	4.3	4.7	3.6	4.1	3.2	3.4	3.7	3.8	4.0	3.4	3.5

Figure 8. Data Normalization Matrix.

Step 3: Determine the extreme values.

Two levels of absolute difference:

$$|E_{nk}| = |E_{0k} - E_{jk}| \quad (3)$$

E_{nk} represents the difference between two extreme values in the matrix, E_{0k} represents the smallest number in the matrix, and E_{jk} represents the largest number in the matrix.

Maximum difference:

$$\max(E_{nk}) = \max_{j=1}^n \max_{k=1}^m |E_{0k} - E_{jk}| \quad (4)$$

Minimum difference:

$$\min(E_{nk}) = \min_{j=1}^n \min_{k=1}^m |E_{0k} - E_{jk}| \quad (5)$$

Step 4: Calculation of correlation coefficients

The correlation coefficient represents the correlation between the comparison series and the reference series at a certain value. The correlation coefficient is calculated according to Equation (6), in which the smaller ρ is, the stronger the difference between the correlation coefficients [19].

$$\zeta_{Jk} = \frac{\min_j \min_k |E_{0k} - E_{Jk}| + \rho \max_j \max_k |E_{0k} - E_{Jk}|}{|E_{0k} - E_{Jk}| + \rho \max_j \max_k |E_{0k} - E_{Jk}|} \quad (6)$$

Here, ρ is the resolution factor and takes the value of 0.5.

Step 5: Calculate the correlation.

Calculate the average value of the correlation coefficient between each index and the reference series at a certain value [20].

$$r_{0J} = \frac{\sum_{k=1}^m \zeta_{Jk}}{m} \quad (7)$$

Here, ζ_{Jk} is the correlation between the comparison series and the reference series; m is the number of evaluation indicator.

Step 6: Use the AHP method to calculate subjective weights.

(1) Establishment of the judgement matrix.

The AHP method can decompose the complex problem into a number of basic units, and group them according to their mutual dominance, forming an orderly progressive hierarchical structure [18], using a 9-point system for the evaluation of the indexes of the two-two comparative scoring. The scoring standards are shown in Table 9, the establishment of the judgement matrix being $A' = \{A_{1m}, A_{2m}, A_{3m}, \dots, A_{nm}\}$.

$$A' = \begin{matrix} A_{11}, A_{12}, A_{13}, \dots, A_{1n} \\ A_{21}, A_{22}, A_{23}, \dots, A_{2n} \\ A_{31}, A_{32}, A_{33}, \dots, A_{3n} \\ A_{41}, A_{42}, A_{43}, \dots, A_{4n} \end{matrix} \quad (8)$$

Table 9. One to nine scaling method.

Scale	Implication
1	by comparison, the two elements are equally important
3	by comparison, one element is slightly more important than the other
5	by comparison, one element is obviously more important than the other
7	by comparison, one element is strongly more important than the other
9	by comparison, one element is extremely more important than the other
2, 4, 6, 8	the middle value between each of the above two adjacent scales

(2) Calculation of relative weights.

The scalar product of each row is calculated according to Equations (9) and (10), and then its geometric mean is determined.

$$a_J = \sqrt[k]{M_J} (J = 1, 2, 3, \dots, n) \quad (9)$$

$$W_J = \frac{a_J}{\sum_{j=1}^k a_j} \quad (10)$$

(3) Calculate the maximum characteristic value of each evaluation index [19].

$$\lambda_{\max} = \frac{1}{n} \sum_{j=1}^n \left(\sum_{j=1}^n a_{jn} a_j / a_j \right) \quad (11)$$

(4) Consistency test.

$$CI = \frac{\lambda_{max} - r}{n - 1} \quad (12)$$

$$\text{CR} = \frac{\text{CI}}{\text{RI}} \quad (13)$$

Step 7: Entropy weighting method to calculate objective weights.

The assessment indicators in the matrix are normalized according to Equation (14):

$$P_{J_k} = \frac{b_{J_k} - \min b_{J_k}}{\max b_{J_k} - \min b_{J_k}} \quad (14)$$

(2) Determine the entropy of each evaluation index.

Calculate the entropy value of the kth indicator according to Equation (15):

$$Y_k = k \sum_{l=1}^m P_{Jk} \ln P_{Jk}, k = 1/\ln m \quad (15)$$

(3) Calculate entropy weights.

$$\omega_k = \frac{1 - Y_k}{n - \sum_{k=1}^n Y_k} \quad (16)$$

Calculate the average of the correlation coefficients between each indicator and the reference series at a certain value according to Equation (17):

$$r'_{0J} = \frac{\sum_{k=1}^m \zeta_{Jk}}{m} \quad (17)$$

The gray weighted correlation is calculated and ranked according to Equation (18), and the top-ranked scheme is the preferred scheme. The indicators with a high weight and low score in the design scheme need to be optimized to a high degree [19].

$$r_{0J} = \frac{\sum_{k=1}^m \omega_k \zeta_{Jk}}{m} \quad (18)$$

The optimal solution of each assessment index is set as the reference sequence, i.e., $(E_0) = (5, 5)$, and the value taken in the correlation coefficient is determined as 0.5.

The smaller the ρ in the formula for calculating the correlation coefficient, the greater the discriminative power [20]. The results of the correlation coefficient calculation are shown in Table 10. The correlation results and their ranking for each functional module solution are shown in Table 11. The subjective weighting table calculated by AHP is shown in Table 12. The table of target weights calculated using entropy weight theory is shown in Table 13.

Table 10. Calculation results of the correlation coefficient of each functional module scheme.

Backrest 1	Backrest 2	Backrest 3	Lower Limb1	Lower Limb2	Lower Limb3	Handrail 1	Handrail 2	Handrail 3
0.691	0.724	0.802	0.639	0.794	0.613	0.692	0.732	0.731
0.765	0.697	0.647	0.753	0.667	0.712	0.669	0.809	0.639
0.848	0.860	0.771	0.743	0.710	0.712	0.673	0.772	0.673
0.715	0.695	0.545	0.643	0.636	0.686	0.727	0.789	0.749
0.766	0.762	0.714	0.736	0.653	0.720	0.658	0.783	0.773
0.742	0.744	0.744	0.793	0.764	0.679	0.641	0.815	0.745
0.639	0.678	0.590	0.590	0.686	0.717	0.769	0.745	0.746
0.666	0.742	0.707	0.676	0.710	0.725	0.666	0.687	0.794
0.756	0.827	0.650	0.607	0.697	0.697	0.861	0.784	0.859
0.675	0.713	0.734	0.659	0.741	0.745	0.733	0.738	0.691
0.714	0.713	0.709	0.739	0.689	0.684	0.685	0.812	0.729
0.753	0.762	0.721	0.602	0.718	0.743	0.761	0.745	0.831
0.788	0.715	0.686	0.702	0.714	0.751	0.710	0.724	0.803
0.713	0.800	0.725	0.743	0.808	0.711	0.670	0.736	0.706
0.876	0.778	0.723	0.729	0.654	0.693	0.822	0.768	0.760
0.735	0.593	0.627	0.767	0.626	0.559	0.571	0.810	0.761
0.866	0.608	0.718	0.692	0.708	0.728	0.687	0.814	0.690
0.682	0.687	0.646	0.758	0.610	0.629	0.613	0.683	0.651
0.732	0.812	0.720	0.720	0.708	0.729	0.693	0.750	0.779
0.663	0.846	0.652	0.718	0.703	0.661	0.710	0.840	0.679
0.591	0.748	0.720	0.603	0.698	0.646	0.692	0.676	0.792
0.774	0.708	0.762	0.624	0.728	0.642	0.615	0.850	0.720
0.692	0.729	0.680	0.588	0.803	0.667	0.717	0.680	0.801
0.631	0.833	0.762	0.582	0.756	0.701	0.759	0.797	0.893

Table 11. Relevance results of each functional module solution and its ranking.

Programs	Backrest 1	Backrest 2	Backrest 3	Lower Limb1	Lower Limb2	Lower Limb3	Handrail 1	Handrail 2	Handrail 3
Relevance Ranking	0.728 2	0.741 1	0.698 3	0.684 3	0.708 1	0.690 2	0.700 3	0.764 1	0.750 2

Table 12. Table of subjective weights calculated by AHP.

Primary Indicators	Secondary Indicators	Contrast Matrix				Eigenvector	Subjective Weights	Amax	CR
E1	E11	1	3	1/3	3	1.022	0.255	4.162	0.061
	E12	1/3	1	1/3	3	0.606	0.152		
	E13	3	3	1	7	2.105	0.526		
	E14	1/3	1/3	1/7	1	0.267	0.067		
E2	E21	1	3	1/3	3	1.102	0.275	4.249	0.094
	E22	1/3	1	1/2	3	0.725	0.181		
	E23	3	2	1	5	1.867	0.467		
	E24	1/3	1/3	1/5	1	0.306	0.076		

Table 12. Cont.

Primary Indicators	Secondary Indicators	Contrast Matrix				Eigenvector	Subjective Weights	$\Lambda_{\max}$	CR
E3	E31	1	1/3	1/2	1/3	0.437	0.109	4.261	0.098
	E32	3	1	1/3	1/2	0.792	0.198		
	E33	2	3	1	1/2	1.171	0.293		
	E34	3	2	2	1	1.599	0.400		
E4	E41	1	1/3	1/2	3	0.728	0.018	4.215	0.081
	E42	3	1	3	3	1.894	0.474		
	E43	2	1/3	1	3	0.989	0.247		
	E44	1/3	1/3	1/3	1	0.388	0.097		
E5	E51	1	1/2	1/3	1/2	0.422	0.105	4.215	0.081
	E52	2	1	1/3	1/2	0.681	0.170		
	E53	3	3	1	1/2	1.282	0.320		
	E54	3	2	2	1	1.615	0.404		
E6	E61	1	1/2	3	3	1.237	0.309	4.143	0.054
	E62	2	1	3	3	1.74	0.435		
	E63	1/3	1/3	1	1/2	0.423	0.106		
	E64	1/3	1/3	2	1	0.6	0.150		
E7	E11	1	3	1/3	3	1.022	0.255	4.121	0.046
	E12	1/3	1	1/3	3	0.606	0.152		
	E13	3	3	1	7	2.105	0.526		

Table 13. Table of objective weights calculated by entropy weight theory.

Evaluation Metrics	Information Entropy Value e	Redundancy Degree d	Objective Weights
E11	0.362	0.638	0.385
E12	0.696	0.304	0.183
E13	0.599	0.401	0.242
E14	0.599	0.314	0.190
E21	0.361	0.639	0.408
E22	0.700	0.300	0.192
E23	0.625	0.375	0.240
E24	0.750	0.250	0.160
E31	0.761	0.239	0.179
E32	0.700	0.300	0.225
E33	0.580	0.420	0.315
E34	0.627	0.373	0.280
E41	0.700	0.300	0.148
E42	0.003	0.997	0.493
E43	0.482	0.518	0.256
E44	0.793	0.207	0.103
E51	0.761	0.239	0.159
E52	0.676	0.324	0.215
E53	0.432	0.568	0.378
E54	0.627	0.373	0.248
E61	0.432	0.568	0.321
E62	0.363	0.637	0.360
E63	0.761	0.239	0.135
E64	0.674	0.326	0.184

The combined weights were calculated using Equation (19), and the results are shown in Table 14.

$$\omega = \tau\omega_1 + (1 - \tau)\omega_2 \quad (19)$$

Table 14. Calculation results of comprehensive weights.

E ¹	E11	E12	E13	E14
C ¹	0.320	0.167	0.384	0.128
E	E21	E22	E23	E24
C	0.342	0.187	0.353	0.118
E	E31	E32	E33	E34
C	0.144	0.212	0.462	0.340
E	E41	E42	E43	E44
C	0.083	0.483	0.252	0.100
E	E51	E52	E53	E54
C	0.132	0.116	0.349	0.326
E	E61	E62	E63	E64
C	0.315	0.398	0.120	0.167

¹ E for Evaluation Indicators; C for Composite Weights.

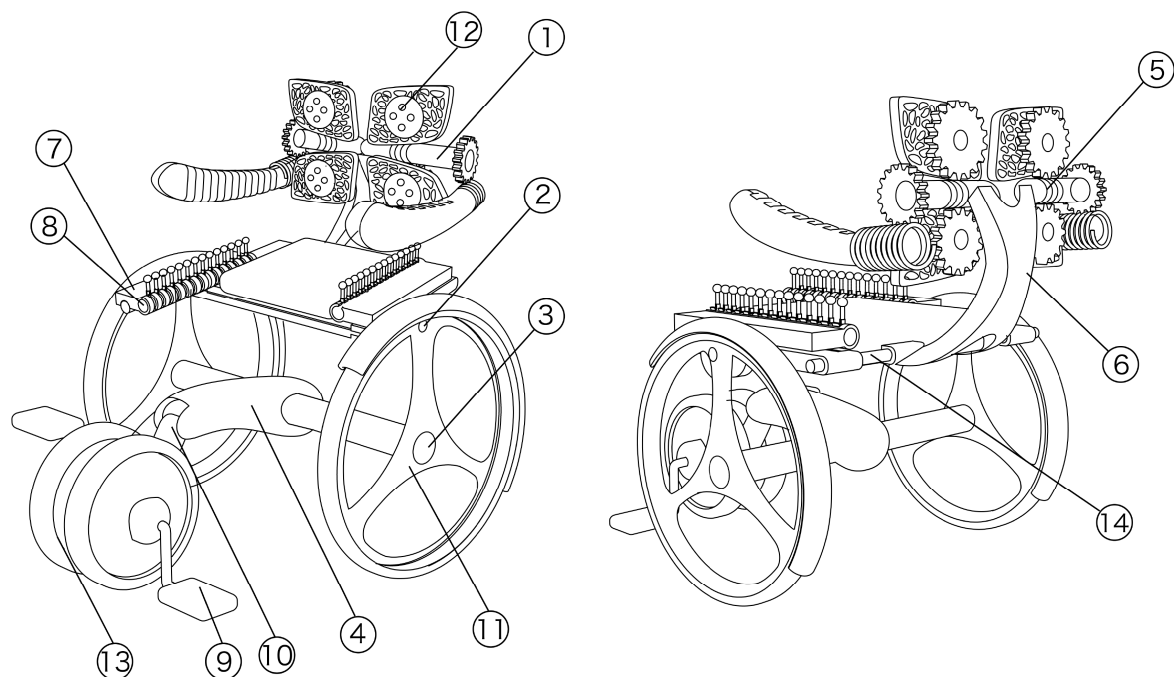
The gray weighted correlations were calculated according to Equation (18) and ranked according to the results, which are shown in Table 15.

Table 15. Grey integrated weight correlation and its ranking.

Programs	Backrest 1	Backrest 2	Backrest 3	Lower Limb1	Lower Limb2	Lower Limb3	Handrail 1	Handrail 2	Handrail 3
Relevance Ranking	0.182 2	0.188 1	0.177 3	0.171 3	0.178 1	0.173 2	0.175 3	0.192 1	0.185 2

3. Results

As shown in Figures 9–11 the functional programs are sorted and combined into three different rehabilitation chairs in the overall program.

**Figure 9.** Rehabilitation chair combination program 1.

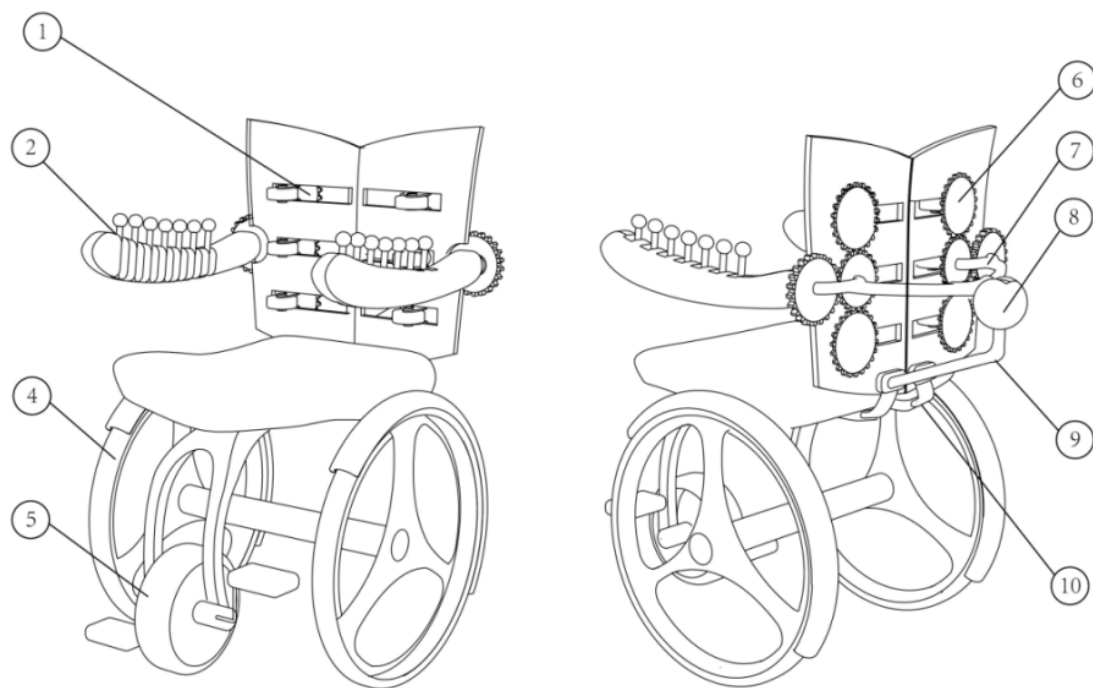


Figure 10. Rehabilitation chair combination program 2.

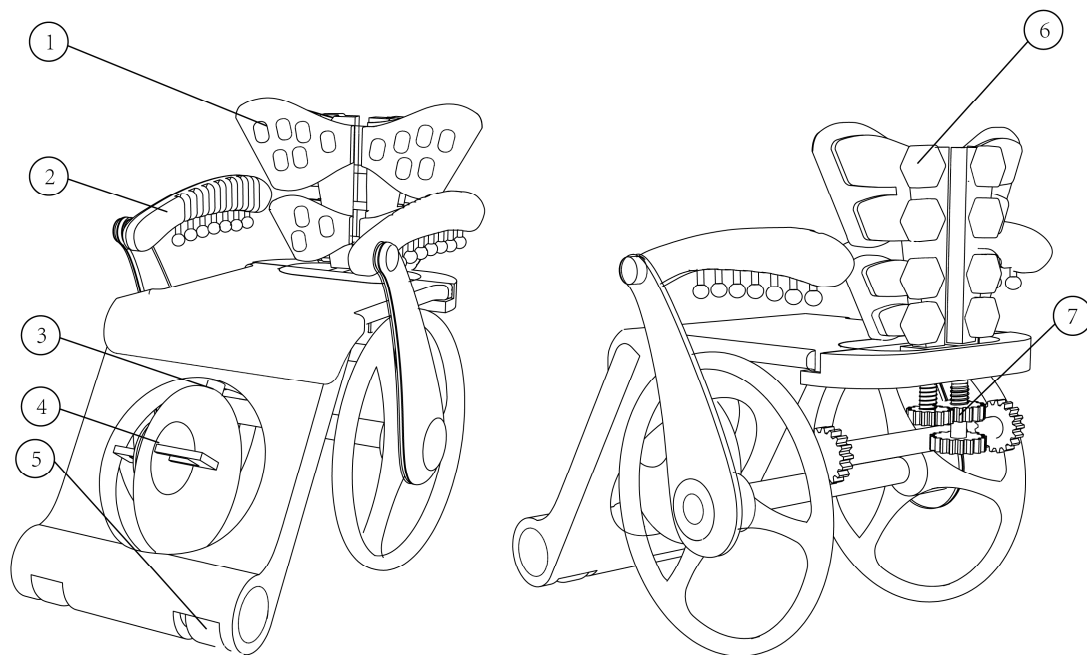


Figure 11. Rehabilitation chair combination program 3.

The first rehabilitation chair combines backrest 2, handrail 2, and lower limb 2 as shown in Figure 9. 1 is and 5 is the worm gear that controls the massage back. 2 is the fixed wheel swivel connector. 3 is the wheel swivel axle. 4 is the leg lift component. 6 is the backrest and seat surface connector. 7 is the leg tap. 8 is the leg tap axle. 9 is the foot pedal. 10 is the connector. 11 is the wheel. 12 is the massage ball. 13 is the treadle wheel. 14 is the backrest swivel axle.

The backrest scheme consists of four pieces with a gear behind each piece, which rotates to complete the back massage function. The user controls the worm gears by rotating the armrests. When the user rotates the armrests, the two worm gears are combined with

two worm gears (1) to form a worm gear structure. By rotating the armrest, joints such as arms and wrists can be exercised. In addition, the user pushes the armrest outward to achieve the chest expansion posture that can be performed to expand the chest exercise. The backrest massage balls can be disassembled to increase the number. By actively massaging the back, it facilitates the rehabilitation effect of the upper limbs and the back. The backrest connecting member 6 connects two worm wheels on the back of the backrest and the back of the seat surface, respectively, and the back of the seat surface is designed with a connecting shaft 14 to achieve the backrest's forward and backward rotation. The leg-knocking structure is provided with a number of wooden leg-knocking heads, a connecting shaft 8, and a plate mounting sleeve 7, the connecting shaft 8 being embedded in the plate mounting sleeve 7, the wooden leg-knocking heads being connected to the connecting shaft 8, and the rotation 8 controlling the wooden leg-knocking heads to knock on the legs of the user. The connecting shaft 8 is removably connected to the plate mounting sleeve 7, and the wooden leg knocker head is also removably connected to the connecting shaft 8. The connecting shaft 8 is telescopically adjustable. The foot structure includes a driving wheel 13, a telescopic connector 10, a lifting structure 4 and a foot pedal 9, the connector 2, and the inner wheel 11 being fixed on both sides of the seat. The foot pedal structure drives the wheels 11 to achieve movement.

The second solution consists of the second-ranked functional solution as shown in Figure 10. 1 is the massage roller. 2 is the armrest knockout structure. 4 is the wheel. 5 is the treadle wheel. 6 is the gear. 7 is the gear connector. 8 and 9 are the backrest and seat surface connectors. 10 is the seat surface connector.

The gears are controlled by turning the armrests, which in turn drive the rollers for massage. The gear at the back of the armrest and the gear at the back of the backrest are connected by component 8, and the control of the armrest to move outward can achieve the chest expansion movement. The inner shaft 2 of the armrest is connected to a row of massage balls, and the knocking head can be retracted, which can realize the function of leg knocking. The foot wheel 5 is connected to the bottom of the seat surface, which can control the height of the seat surface, and after adjusting the seat surface to a higher level, the armrest can carry out percussion massage on the legs. The seat surface and the backrest structure are connected through the assembly 10.

The backrest massage structure in Scheme III consists of hexagonal parts 6 that control the backrest, the surface of the backrest is massage roller 1, and the armrest can be moved forward and backward or rotated to control the massage height and massage angle, so as to achieve the purpose of adjusting the position of the leg patting massage. Gear combination 7 is a structure to control the left and right opening and closing of the backrest. The foot pedal is embedded in the round plate 4, and the round plate 4 rotates with the shaft 3, which can be stowed to form a complete leg support plate. After rotating out the foot pedal, the lower limbs can complete the foot movement, as shown in Figure 11.

1 is the roller. 2 is the armrest knockout structure. 3 is the footrest swivel shaft. 4 is the footrest. 5 is the universal wheel. 6 is the backrest control structure. 7 is the gear and worm gear connectors.

The validation evaluation of the three combined solutions was carried out by distributing 187 questionnaires, and the eight dimensions of usability, interest, practicality, safety, comfort, aesthetics, innovation, and ergonomics of the product were rated using the Richter Scale method. Through the radar chart to connect the indicator feature vectors, the resulting area can reflect the effectiveness of the medical rehabilitation chair design scheme. The results will be plotted as a radar chart as shown in Figure 12, from which it can be seen that the largest area is red, on behalf of the first-ranked program, green for the second-ranked program combinations, and yellow for the third-ranked program combinations, which is more in line with the multi-level evaluation of the resulting functional program ideas with a reference value.

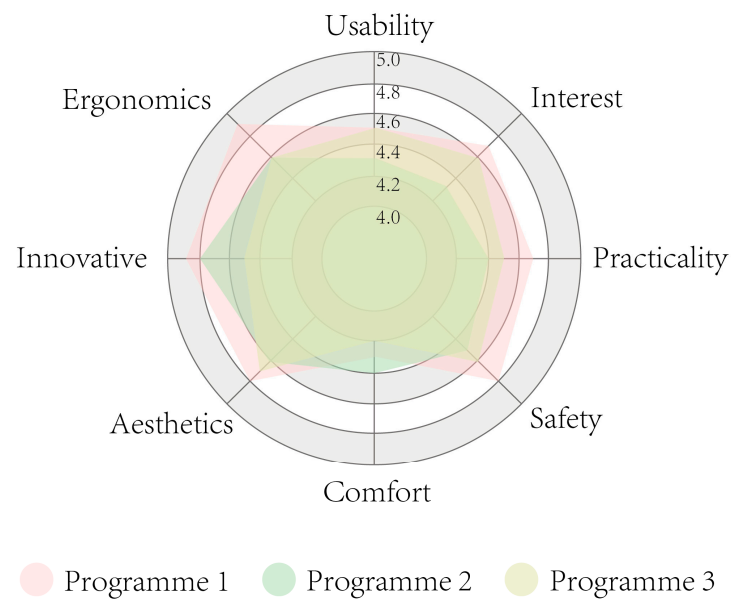


Figure 12. Radar chart for program evaluation.

Finally, the overall shape of the rehabilitation chair was designed, and the product is in blue and white colors as shown in Figure 13. This rehabilitation chair solves the problems of other products such as unfocused massage range and expensive cost. This rehabilitation chair can achieve stretching function, chest expansion movement, arm lifting movement, flexion and extension movement, massage function, and foot pedal function in order to improve the patient's rehabilitation initiative.



Figure 13. Optimal solution design for rehabilitation chairs.

4. Discussion

This paper combines the AHP theory, the entropy weight method, the grey correlation analysis method, and the function matrix method, overcomes the problem that purely using individual methods will fall into incomplete decision-making, and analyses the relationship between user requirements, technical features, and functional components in the design of rehabilitation chairs through the combination of qualitative and quantitative methods, realizing mutual transformation between the three. The grey correlation analysis method applies first-order linear differential equations to establish mapping relationships, which makes it difficult to quantify the relationship between user requirements and design features. When using this method, this part needs to be supplemented. In this paper, a functional technology matrix is used to compensate for the lack of quantification of user requirements.

There are examples of design methods used in this paper that have been applied in other areas. For example, Yanli Yuan designed a manual wheelchair using the AHP and Kano Model to improve design efficiency [21]. Mahmoud Z. Mistarihi designed a wheelchair design with nested seat backs and armrests using QFD modelling and the AHP method and, as in this study, the functional design was based around user requirements [22]. Han Yue proposed an improved AHP method for product design evaluation, which can provide a reference for the design of home healthcare products for the elderly [11]. Yanling Wu integrated AD, TRIZ, fuzzy, and grey correlation analysis into the product design process and proved that the integrated method is effective through case studies in both the manufacturing and chemical industries [23]. On the other hand, we reviewed the research related to rehabilitation wheelchairs. Muhammad Zia Ur Rahman designed a modular wheelchair where the user only needs to control it using their body weight and muscle power to perform sitting and standing transitions. This structure is beneficial for the rehabilitation of human lower limbs [24]. Rosnani Ginting also designed a multifunctional wheelchair using questionnaire method research, before using the QFD method and Nigel Cross Method to deepen the solution [25]. Gazi Akgun concluded that exoskeletons are either too complex or too simple with limited functionality to combine active and passive rehabilitation and therefore developed an adaptive hand rehabilitation robot [26]. Constantinos Mavroidis proposed a portable rehabilitation device that can improve the rehabilitation process [27]. Su-Hong Eom proposed a human joint-like device for connecting and rehabilitating lateral and bilateral movement scenarios for autonomous rehabilitation of upper limb hemiplegic patients [28]. Han Jianhai designed an active rehabilitation training system based on virtual reality technology, aiming to increase patients' interest in rehabilitation at a low price, but more severe patients are unable to control their limbs autonomously and need hardware products for more reliable rehabilitation [29]. Tian Shi presented a numerical method for solving the active set conjugate gradient method, which is feasible and effective for MPC lower limb rehabilitation robots trained in both passive and active rehabilitation [30]. Khaled M. Goher presented the design and development of a prototype of a reconfigurable wheelchair for rehabilitation and self-assistance, where the user can use an adjustable back support with two linear actuators to adjust the posture of the upper body [31]. Xin Zhang designed an exoskeleton lower limb rehabilitation product for the elderly, discussed an adaptive fuzzy control scheme, developed an "active" lower limb training device for the elderly, and implemented a position-tracking controller [32]. Qiaoling Meng proposed a wheelchair-based powered exoskeleton for upper limb rehabilitation [33]. Xinyan Yang designed a manual rehabilitation seat using FBS and QFD models, verifying that the functional design solution based on situational needs is reliable [34].

According to the existing research, it can be found that the research on rehabilitation products is wide-ranging, and most of them are designed using the principles of mathematics and other disciplines. We categorize the rehabilitation product research as follows: in terms of the product form, it includes (1) research on multifunctional rehabilitation products using products such as beds or wheelchairs as a carrier; (2) corrective or training equipment focusing on the human body's local functional rehabilitation; (3) rehabilitation based on digital technology posture research; (4) multimodal-based rehabilitation research. In terms of product design methodology, it mainly includes (1) product design research based on product realization technology and principles; (2) product design research based on medical rehabilitation theory; (3) product design research based on the theory of user behavior, and so on. We reviewed studies related to rehabilitation wheelchairs and found that there are more mature studies on wheelchair structure and function, but fewer studies on active rehabilitation product design. Divanoglou et al. describe active rehabilitation as "the teaching of practical life and social skills from the grassroots level by experienced, active spinal cord injury (SCI) patients (peer mentors) to newly injured individuals or others in need" (p. 545) [35].

This study provides a wheelchair product for user-initiated rehabilitation through questionnaire research and other methods in the hope that it will provide a design reference for related products in terms of structure and function.

5. Conclusions

In this paper, the quantitative value of the functional requirements of the rehabilitation chair was obtained by using the questionnaire survey method and the functional-technical matrix method, and a specific functional solution was designed based on the results. The engineering design problem between functions is effectively solved. The rehabilitation chair can provide patients with the functions of back acupressure, upper limb exercise, leg acupressure tap, and leg strength exercise. In the selection of functional solutions, the AHP method, entropy weight method, and grey correlation analysis are combined to evaluate the functional solutions, and then the combination design is carried out according to the sorting results. The Likert scale method is used to verify whether the three kinds of rehabilitation wheelchairs are in line with the results of the multilevel evaluation. After verification, the results are consistent with the multilevel evaluation results.

This paper contributes in the following aspects:

- (1) In order to reduce the subjective bias in decision-making, the objectivity of the design strategy is improved by combining the functional technology matrix, grey correlation analysis, AHP, and entropy value method.
- (2) Using the functional matrix method to quantitatively analyze the obtained user research results, thus achieving the decomposition of the design functions of the rehabilitation chair and satisfying the rehabilitation training needs of the researched users as much as possible.

The research in this paper has the following significance:

- (1) In the context of global energy constraints and serious material waste, it is crucial to design rehabilitation products that are sustainable in use, green, low-carbon, and environmentally friendly.
- (2) The research ideas and methods in this paper can provide a reference for related product design and can also be used to try to solve the sustainable design problems of other industrial products.
- (3) Combining statistics, decision science, mathematics, and design is an important research direction for future product design.

It is worth noting that this study makes up for the shortcomings of the questionnaire research, which is biased towards subjective judgement, and quantifies the functions by using the Functional Technology Matrix. And the multilevel evaluation method used in this paper was verified using program design and the five-level evaluation method, and the results were reliable. However, there are only three design options for each rehabilitation function in this study, which is a small number and can only be used as an illustrative design methodology study. More options can be designed in the future to improve the aesthetics, practicality, and rehabilitation effect of the rehabilitation chair. In addition, due to the large number of structures involved in this study, they need to be further analyzed in subsequent studies to make them meet the design expectations.

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